

Task 9

Cybersecurity Internship – Phase 1 (Task 10/10)

Task Title: Creating a Firewall with iptables or UFW

Date Completed: August 26, 2025

Platform: Ubuntu 22.04 (Linux)

Tool Used: UFW (Uncomplicated Firewall)

Objective

The objective of this task was to understand and implement basic firewall rules using UFW, a user-friendly firewall management tool on Linux. This task covered the creation, modification, and inspection of firewall rules to control traffic by IP address, port, and protocol.

Setup & Procedure

1. UFW Installation and Activation:

```
sudo apt install ufw  
sudo ufw enable
```

2. Set Default Policies:

Deny all incoming connections by default and allow all outgoing traffic:

```
sudo ufw default deny incoming  
sudo ufw default allow outgoing
```

3. Allowing Specific Services:

- Allow SSH for remote administration:

```
sudo ufw allow ssh
```

- Allow HTTP (port 80) for web traffic:

```
sudo ufw allow 80/tcp
```

- Allow HTTPS (optional):

```
sudo ufw allow 443/tcp
```

4. Blocking a Suspicious IP:

- Block access from IP `192.168.1.50` (example of a potentially malicious actor):

```
sudo ufw deny from 192.168.1.50
```

5. Viewing Firewall Status:

- To review all active rules:

```
sudo ufw status verbose
```

Findings

- All default-deny policies were correctly enforced.
- SSH and HTTP access were permitted as intended.
- The suspicious IP was successfully blocked from reaching the system.
- UFW provided a simplified interface while still allowing powerful rule control.
- Verified rules were persistent across reboots.

Sample Output

Status: active

To	Action	From
--	-----	----
22	ALLOW	Anywhere
80/tcp	ALLOW	Anywhere
443/tcp	ALLOW	Anywhere
Anywhere	DENY	192.168.1.50

Learning Outcomes

- Gained hands-on experience configuring and managing a host-based firewall.
- Learned the difference between **incoming** and **outgoing** traffic controls.
- Understood how to allow or deny traffic based on **port, protocol, and IP address**.
- Explored real-world security use cases like **IP blocking** and **service whitelisting**.
- Learned how firewalls act as the first line of defense against unauthorized access.

Security Insights

- **Firewalls** mitigate common network-based threats such as port scanning, unauthorized access, and brute-force login attempts.
- Blocking unused ports and limiting allowed IP ranges reduces the system's attack surface.
- Even beginner tools like UFW can be highly effective when configured correctly.

Interview Questions & Answers

1. What is a firewall?

A firewall is a security system that monitors and controls incoming and outgoing network traffic based on predefined rules.

2. What's the difference between iptables and UFW?

`iptables` is a powerful but complex command-line firewall tool, whereas `UFW` is a simplified frontend designed to make firewall management easier for beginners.

3. What does `sudo ufw default deny incoming` mean?

It sets the default rule to block all incoming traffic unless explicitly allowed.

4. How can you allow only SSH and block everything else?

Set default policy to deny incoming traffic, then allow only SSH:

```
sudo ufw default deny incoming
sudo ufw allow ssh
```

5. What are INPUT, OUTPUT, and FORWARD chains in iptables?

- **INPUT:** Handles incoming traffic.
- **OUTPUT:** Handles outgoing traffic.
- **FORWARD:** Handles traffic being routed through the host.

6. How do stateful firewalls differ from stateless?

Stateful firewalls track the state of active connections, allowing return traffic from established sessions, while stateless firewalls filter each packet individually.

7. How do you permanently save iptables rules?

Use `iptables-persistent` or `netfilter-persistent` :

```
sudo apt install iptables-persistent
sudo netfilter-persistent save
```

8. What risks do firewalls help mitigate?

Unauthorized access, malware spread, port scanning, and data exfiltration attempts.

9. Can you configure a firewall to allow only specific IPs?

Yes. For example, with UFW:

```
sudo ufw allow from 192.168.1.10
```

10. How can attackers bypass poorly configured firewalls?

Through open ports, misconfigured rules, tunneling techniques (like SSH or VPNs), or by exploiting firewall exceptions.